

Digital Parking Permit System – Final Report

Course: SER216 **Student Name:** Raxmova Gulira'no **Date:** 4 May 2026 **GitHub Repository:** [Your GitHub Link]

Task 1: Requirements

Functional Requirements

FR1 – Vehicle Registration Students and staff can register a vehicle by entering their ID and vehicle plate number into the system.

FR2 – Permit Request Registered users can submit a parking permit request by selecting an available parking zone.

FR3 – Zone Availability Check Users can view real-time availability of parking zones before submitting a permit request.

FR4 – Violation Logging Security staff can verify the validity of a permit and log parking violations into the system.

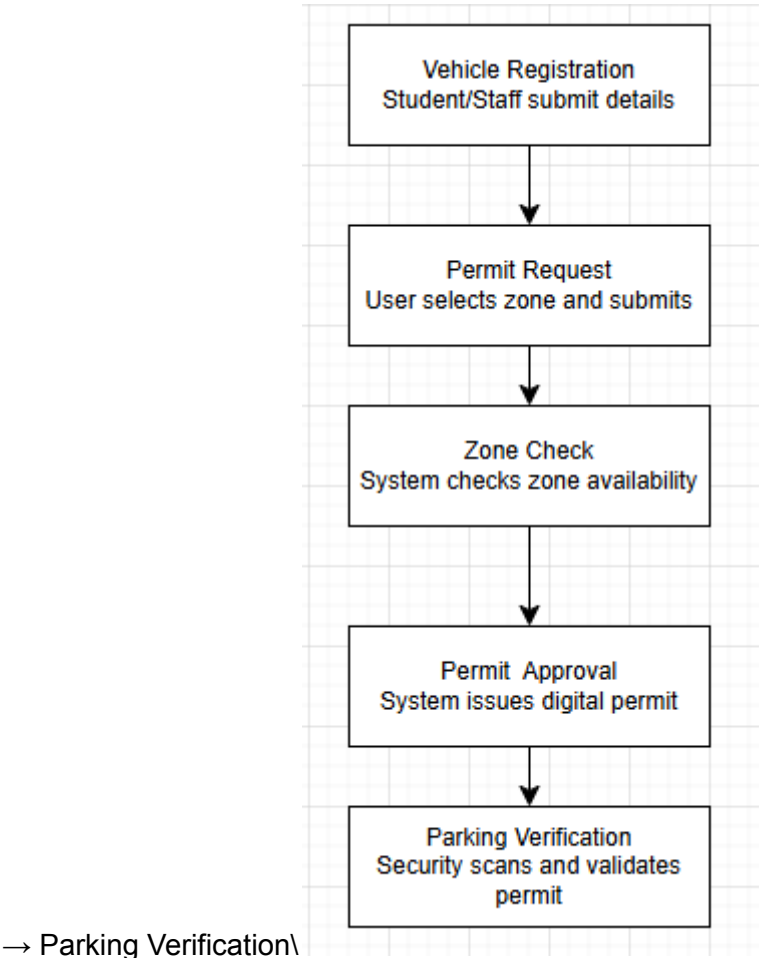
Non-Functional Requirements

NFR1 – Performance The system must display permit status and zone availability within 3 seconds under normal network conditions.

NFR2 – Security All user data and permit records must be stored securely, with access restricted by user role (student, staff, security).

Task 2: Workflow Diagram

Workflow steps: Vehicle Registration → Permit Request → Zone Check → Permit Approval



Task 3: Test Case Table

Test ID	Feature	Input / Data	Expected Result	Test Type
TC01	Vehicle Registration	Valid student ID + plate number	Vehicle registered successfully	System
TC02	Permit Request	Registered vehicle + available zone	Permit request submitted	System
TC03	Register Vehicle	Student submits complete form	System confirms registration	Acceptance
TC04	Permit Approval Flow	User requests permit, admin approves	User receives approved permit	Acceptance
TC05	Permit Approval (after fix)	Previously approved permit data	Permit still shows as valid	Regression

TC06	Zone Check (after update)	Query zone availability post-update	Correct zone status returned	Regression
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Task 4: Defect Analysis

Defect: *"A valid parking permit is shown as expired during security verification."*

Item	Answer
Defect Category	Logic / Data Error
Severity	High
Priority	High
Possible Root Cause	Permit expiry date comparison uses wrong date format or timezone mismatch
Regression Test	Verify all valid permits display correct status after applying the fix

Task 5:

https://github.com/Lilys911/SER216_Final_Project_Gulira-no